Traffic Sign and Signal Recognition in Bad Conditions

Fuzzy-logic image enhancement, neural network classification and YOLO detection

One project for Deep Learning · Image Processing and Analysis · Neural Networks and Fuzzy Logic

0.819
macro F1 on 203,374 real damaged test signs

97.9%
accuracy on 281,547 traffic-light test images

106
sign and light classes from 7 datasets

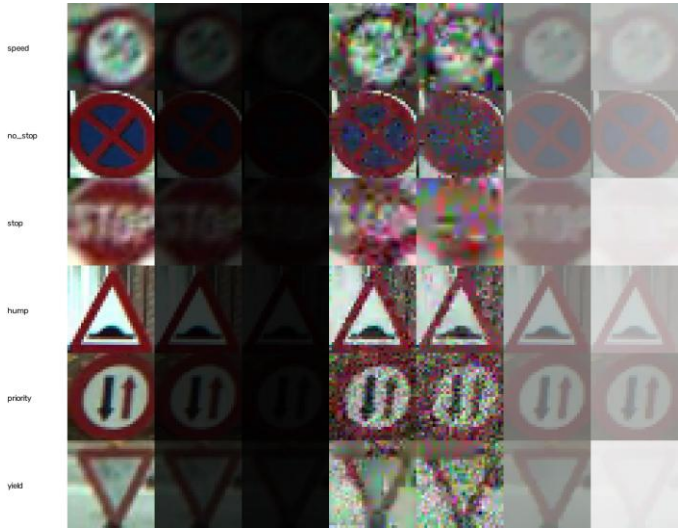
0.786
on worn real signs after wear training (0.597 before)

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Signs must be read in the conditions where vision fails

Problem statement

- Night, haze, rain, snow, noise, blur and worn paint damage the image a driver-assistance camera sees.
- A CNN trained on clean images reads only 0.683 macro F1 on real damaged signs, against 0.819 when trained on damage.
- Can image processing, steered by fuzzy logic, repair images enough to help, and how does that compare with training on damage?

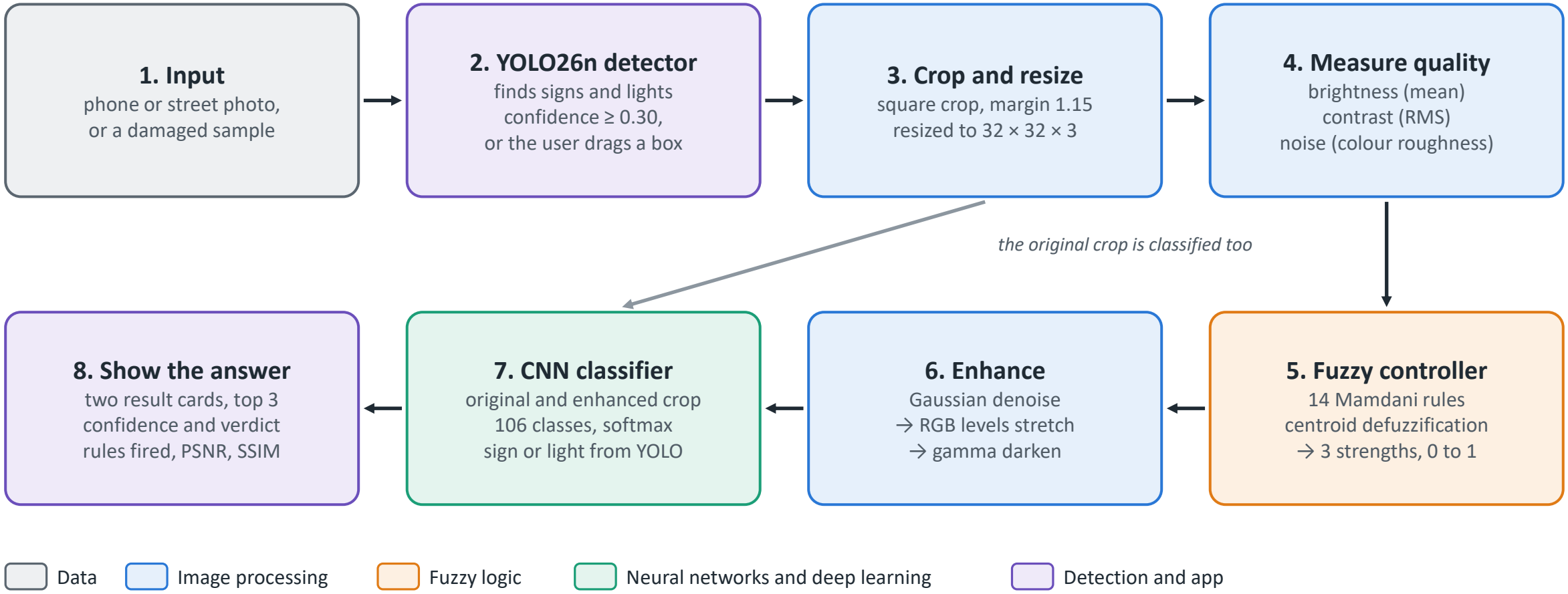


CURE-TSR signs: clean, then darkening, noise and haze at levels 3 and 5

Objectives

1. **Image processing**
   filters, histograms, convolution, measured with PSNR and SSIM
2. **Fuzzy logic**
   a Mamdani controller that chooses denoise, stretch and darken strengths
3. **Neural networks**
   MLP, feature MLP and backpropagation written by hand
4. **Deep learning**
   CNNs on clean, damaged and enhanced images, 17 to 106 classes, worn signs
5. **Detection and app**
   YOLO finds signs; a Streamlit app shows every step; honest scoring

Solution: measure, enhance with fuzzy logic, then classify



Denoising goes first, because stretching would amplify leftover noise. Colours show which course each step belongs to.

Architecture: four layers, from datasets to the app



Technology stack

Layer	Tools
Language	Python 3.11, virtual environment
Deep learning	PyTorch 2.15 to train, 2.5.1 for the app; RTX 4050 6 GB
Detection	Ultralytics 8.4, YOLO26 nano
Image processing	OpenCV, scikit-image, NumPy
Fuzzy logic	scikit-fuzzy sets, NumPy inference
Evaluation	scikit-learn, pandas, matplotlib
App	Streamlit, streamlit-cropper, Cloudflare tunnel

All 262 automated checks pass.

7 datasets, 51 classifier runs, every test set scored once

Dataset	What it adds	Images used
CURE-TSR	14 signs, 12 damages × 5 levels	2,206,100
LISA	red, yellow, green lights, day and night	41,923 crops
GTSRB	real German sign photos	51,839 crops
Indian set	real Indian sign photos	4,359 crops
Mapillary	street signs, 6 continents, 'other sign'	48,863 crops
Kaggle damaged	faded, rusted, vandalised signs	3,194 photos
Roboflow damaged	close-ups by kind of damage	1,996 photos

51
classifier training runs, 3 seeds each

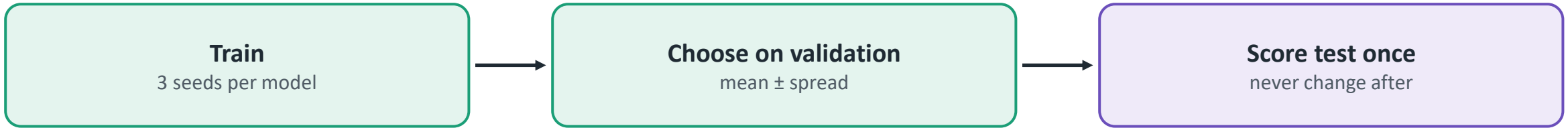
2
YOLO detectors, 9.4 hours of training

1.92 M
training images for the final CNN

106
classes: 17 → 84 → 106

No leakage: splits by whole clip, track or block of frames; near-duplicates removed; test street photos kept out of training.

Clean crops before synthetic damage; Kaggle and Roboflow frames also train YOLO.



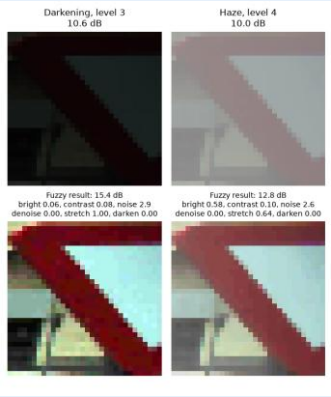
Three courses, one pipeline

Image Processing

34.83 dB

median filter on 2% salt-and-pepper (from 22.16)

- point operations, histogram equalization, CLAHE
- convolution by hand, equal to OpenCV
- new colour-roughness noise measure



Darkening and haze: damaged (top) and after fuzzy enhancement (bottom)

Fuzzy Logic

0.683 → 0.715

real signs F1, clean-trained CNN with fuzzy input

- 3 inputs, 3 outputs, 3 sets each
- 14 Mamdani rules, centroid output
- within 0.0001 of scikit-fuzzy

IF brightness is bright
AND contrast is medium
THEN darken is strong

one of 14 rules: min for AND, max to combine, centroid output

Neural Nets and Deep Learning

5.2×10^{-7}

worst gradient error of the hand-written MLP

- MLP, feature MLP, CNN compared
- CNN: 6 conv layers, batch norm, dropout
- 17 → 106 classes, YOLO, worn signs

816 K
CNN weights

6
conv layers

2
YOLO models

17 → 84 → 106 classes over 4 stages

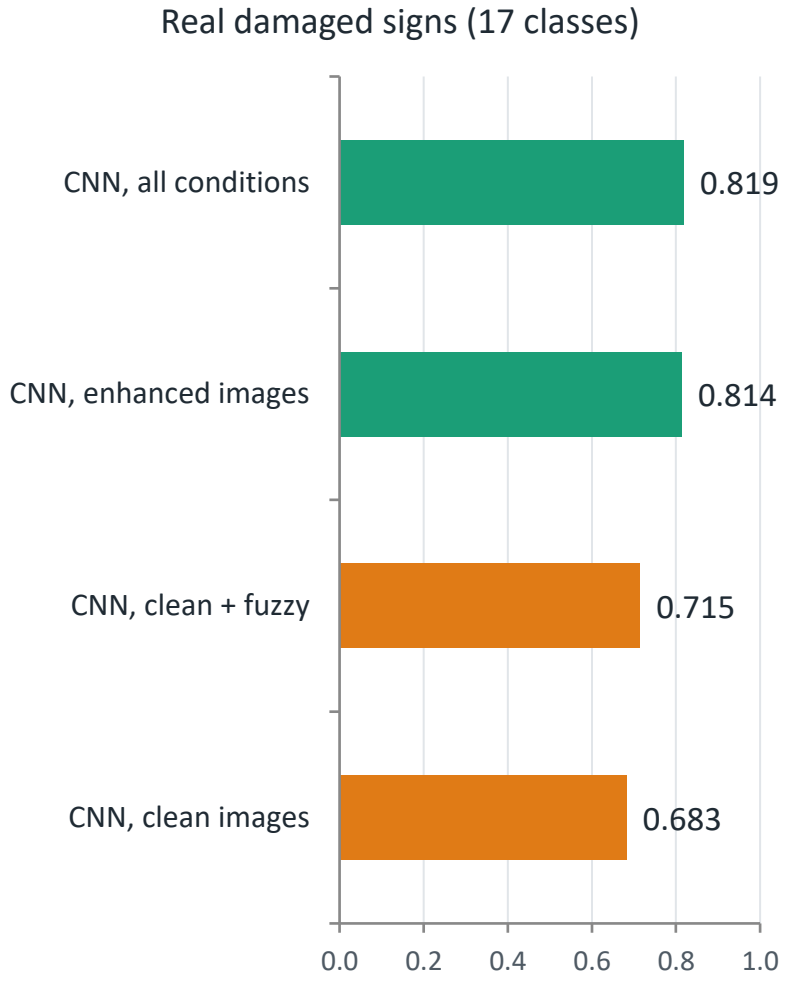
Results: training on damage beats repairing it

Model (test, macro F1)	Classes	Real signs	Simulated	Lights	Worn real
Feature MLP, all conditions	17	0.701	0.825	0.898	–
MLP, all conditions	17	0.733	0.878	0.960	–
CNN, clean + fuzzy input	17	0.715	0.791	0.936	–
CNN, all conditions	17	0.819	0.909	0.980	–
Wide CNN, all conditions	84	0.757	0.881	0.946	–
Global CNN, all conditions	106	0.816	0.888	0.951	0.597
Global CNN + worn signs (app)	106	0.808	0.881	0.959	0.786

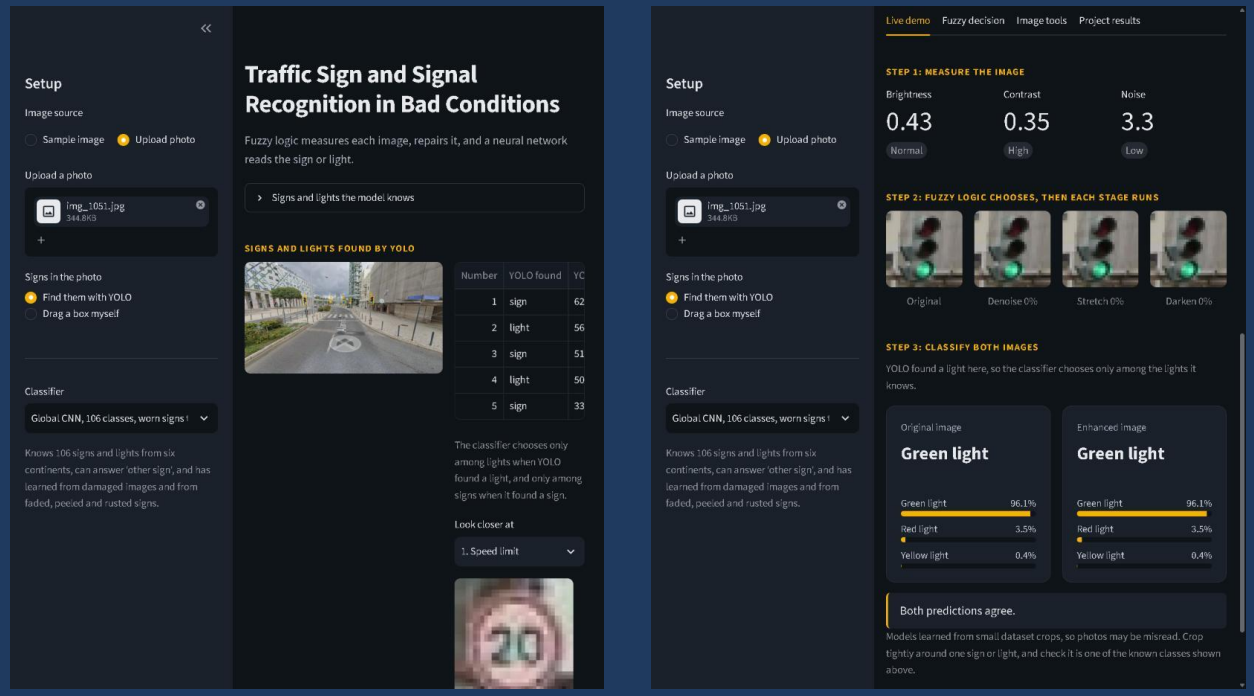
YOLO detector 2, test mAP50: 0.701 signs · 0.680 lights · 0.853 Kaggle damaged · 0.893 Roboflow close-ups (0.003 before fine-tuning)

Best 17-class CNN: 89.8% accuracy on 203,374 real signs, 97.9% on 281,547 lights. Fuzzy enhancement raised PSNR from 17.90 to 18.35 dB.

Worn-sign training: worn real signs 0.597 → 0.786, worn GTSRB 0.629 → 0.867; the usual test sets moved by under 1 point.



A working app, and what we learned



Left: the worn-sign YOLO finds five objects in a street photo. Right: fuzzy stages and both answers, green light 96%.

Conclusions

- Training on damaged images beats repairing them: 0.819 against 0.715 macro F1.
- Fuzzy enhancement still helps every model that never saw damage, and leaves clean images alone.
- Real photos exposed too few classes, close-up lights and worn paint; each fix was measured before it was kept.

Next steps

- Larger input or a pretrained backbone, to read text and numbers
- Stronger sun-bleaching and real photos of old signs
- Night and rain photo tests; YOLO trained on damaged frames

Thank you · Questions?